THE THIRTY-FOURTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

December 1, 1973

A–1. (a) Let ABC be any triangle. Let X, Y, Z be points on the sides BC, CA, AB respectively. Suppose the distances $\overline{BX} \leqq \overline{XC}$, $\overline{CY} \leqq \overline{YA}$, $\overline{AZ} \leqq \overline{ZB}$ (see Figure 1). Show that the area of the triangle XYZ is $\geqq$ (1/4) (area of triangle ABC).

(b) Let ABC be any triangle, and let X, Y, Z be points on the sides BC, CA, AB respectively (but without any assumption about the ratios of the distances $\overline{BX}/\overline{XC}$, etc.; see Figures 1 and 2). Using (a) or by any other method, show: One of the three corner triangles AZY, BXZ, CYX has an area $\leqq$ area of triangle XYZ.

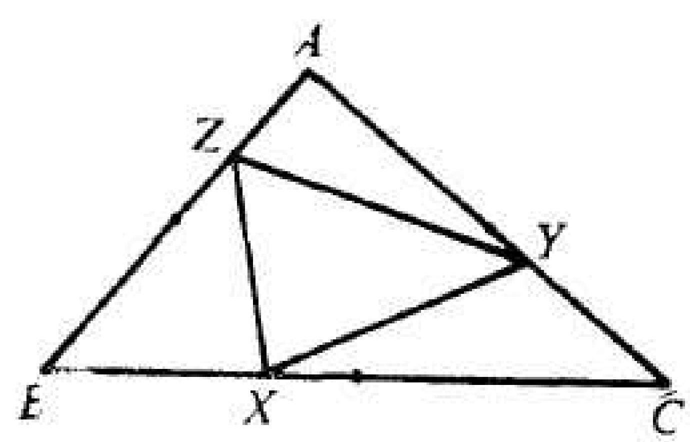

FIG. 1

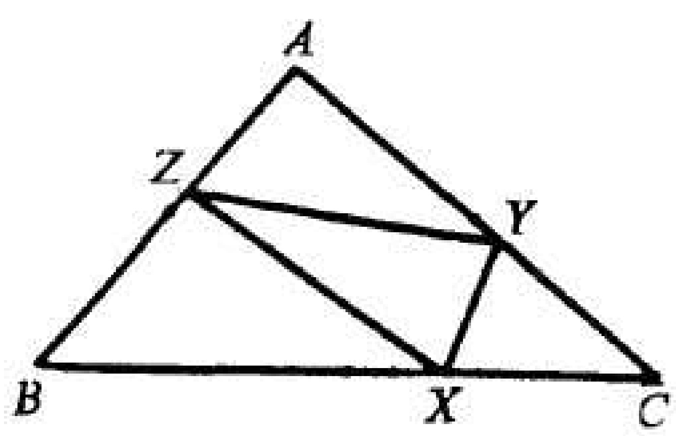

FIG. 2

A–2. Consider an infinite series whose nth term is $\pm (1/n)$, the $\pm$ signs being determined according to a pattern that repeats periodically in blocks of eight. [There are 2^8 possible patterns of which two examples are:

$$+ + - - - - + + ,$$

$$+ - - - + - - - .$$

The first example would generate the series

$$1 + (1/2) - (1/3) - (1/4) - (1/5) - (1/6) + (1/7) + (1/8)$$

$$+ (1/9) + (1/10) - (1/11) - (1/12) - \cdots .]$$

(a) Show that a sufficient condition for the series to be conditionally convergent is that there be four "+" signs and four "−" signs in the block of eight.

(b) Is this sufficient condition also necessary?
[Here "convergent" means "convergent to a finite limit."]

A–3. Let n be a fixed positive integer and let $b(n)$ be the minimum value of

$$k + \frac{n}{k}$$

as k is allowed to range through all positive integers. Prove that $b(n)$ and $\sqrt{4n+1}$ have the same integer part. [The "integer part" of a real number is the greatest integer which does not exceed it, e.g. for π it is 3, for $\sqrt{21}$ it is 4, for 5 it is 5, etc.]

A–4. How many zeros does the function $f(x) = 2^x - 1 - x^2$ have on the real line? [By a "zero" of a function f, we mean a value x_0 in the domain of f (here the set of all real numbers) such that $f(x_0) = 0$.]

A–5. A particle moves in 3-space according to the equations:

$$\frac{dx}{dt} = yz, \quad \frac{dy}{dt} = zx, \quad \frac{dz}{dt} = xy.$$

[Here $x(t)$, $y(t)$, $z(t)$ are real-valued functions of the real variable t.] Show that:

(a) If two of $x(0)$, $y(0)$, $z(0)$ equal zero, then the particle never moves.

(b) If $x(0) = y(0) = 1$, $z(0) = 0$, then the solution is:

$$x = \sec t, \; y = \sec t, \; z = \tan t;$$

whereas if $x(0) = y(0) = 1$, $z(0) = -1$, then

$$x = 1/(t+1), \; y = 1/(t+1), \; z = -1/(t+1).$$

(c) If at least two of the values $x(0)$, $y(0)$, $z(0)$ are different from zero, then either the particle moves to infinity at some finite time in the future, or it came from infinity at some finite time in the past. [A point (x, y, z) in 3-space "moves to infinity" if its distance from the origin approaches infinity.]

A–6. Prove that it is impossible for seven distinct straight lines to be situated in the euclidean plane so as to have at least six points where exactly three of these lines intersect and at least four points where exactly two of these lines intersect.

B–1. Let $a_1, a_2, \ldots, a_{2n+1}$ be a set of integers such that, if any one of them is removed, the remaining ones can be divided into two sets of n integers with equal sums. Prove $a_1 = a_2 = \cdots = a_{2n+1}$.

B–2. Let $z = x + iy$ be a complex number with x and y rational and with $|z| = 1$. Show that the number $|z^{2n} - 1|$ is rational for every integer n.

B–3. Consider an integer $p > 1$ with the property that the polynomial $x^2 - x + p$ takes prime values for all integers x in the range $0 \leqq x < p$. (Examples: $p = 5$ and $p = 41$ have this property.) Show that there is exactly one triple of integers a, b, c satisfying the conditions:

$$b^2 - 4ac = 1 - 4p,$$

$$0 < a \leqq c,$$

$$-a \leqq b < a.$$

B-4. (a) On $[0, 1]$, let f have a continuous derivative satisfying $0 < f'(x) \leqq 1$. Also suppose that $f(0) = 0$. Prove that

$$\left[\int_0^1 f(x)dx\right]^2 \geqq \int_0^1 [f(x)]^3 dx.$$

[Hint: Replace the inequality by one involving the inverse function to f.]

(b) Show an example in which equality occurs.

B-5. (a) Let z be a solution of the quadratic equation

$$az^2 + bz + c = 0$$

and let n be a positive integer. Show that z can be expressed as a rational function of z^n, a, b, c.

(b) Using (a) or by any other means, express x as a rational function of x^3 and $x + (1/x)$. (Display your answer explicitly in a clearly visible form.)
[By a rational function of several variables, we mean a quotient of polynomials in those variables, the polynomials having rational numbers as coefficients, and the denominator being not identically zero. Thus to obtain x as a rational function of $u = x^2$ and $v = x + (1/x)$, we could write $x = (u + 1)/v$.]

B-6. On the domain $0 \leqq \theta \leqq 2\pi$:

(a) Prove that $\sin^2\theta \cdot \sin(2\theta)$ takes its maximum at $\pi/3$ and $4\pi/3$ (and hence its minimum at $2\pi/3$ and $5\pi/3$).

(b) Show that

$$\left| \sin^2\theta \{\sin^3(2\theta) \cdot \sin^3(4\theta) \cdots \sin^3(2^{n-1}\theta)\} \sin(2^n\theta) \right|$$

takes its maximum at $\theta = \pi/3$. (The maximum may also be attained at other points.)

(c) Derive the inequality:

$$\sin^2\theta \cdot \sin^2(2\theta) \cdot \sin^2(4\theta) \cdots \sin^2(2^n\theta) \leqq (3/4)^n.$$